Project Foundation: Predictive Classification of Urban Traffic Congestion in Bangalore

Group 5.1 | John Loyd Arcilla and Sean Mikheal Fulgencio · Oct 29, 2026

1. Project Foundation

The project classifies days of heavy traffic in Bangalore as High Congestion (Congestion Level at or above 80%) or Moderate/Low, so traffic analysts and city planners can flag high-risk conditions before gridlock sets in. It follows the structure of the Group 5.2 project summary: foundation, findings, problem, audience, analytics, then scope and next steps.

Domain and purpose. Urban traffic management. The question is whether volume, speed, incidents, road, weather, and calendar attributes can predict severe congestion across eight Bangalore sectors.

Item	Dataset details
Dataset	Bangalore Traffic Dataset (Banglore_traffic_Dataset.csv), Kaggle, published by Aishwarya Ramachandran
Coverage and size	8,936 rows and 16 raw columns, January 1, 2022 to August 9, 2024 (952 distinct dates)
Unit of analysis	One daily observation at one road or intersection (8 areas, 16 roads)
Target	high_congestion: 1 if Congestion Level is 80 or more (5,557 rows, 62.2%), else 0 (3,379 rows, 37.8%)
Predictors kept (11)	Traffic Volume, Average Speed, Incident Reports, Pedestrian and Cyclist Count, Month, IsWeekend, Area Name, Road/Intersection Name, Weather Conditions, Roadwork and Construction Activity, DayOfWeek
Dropped (9)	Congestion Level (target source), Environmental Impact (exact duplicate of Traffic Volume), Road Capacity Utilization and Travel Time Index (leakage), Public Transport Usage, Traffic Signal Compliance, Parking Usage (noise), Date and Parsed_Date (replaced by calendar features)

Preparation completed. The data has no missing values. Date was parsed into Month, DayOfWeek, and IsWeekend. The cleaned file bangalore_traffic_cleaned.csv (8,936 rows, 12 columns) was exported. A stratified 80/20 split (7,148 train, 1,788 test, class ratio held at 62.2/37.8) and a ColumnTransformer (StandardScaler plus OneHotEncoder, fitted on training data only) are built.

2. Findings Summary

Five EDA findings shape the modeling plan. They describe this dataset only and do not show cause and effect.

1. High congestion is the majority class. 62.2% of rows are High, so a model that always predicts High is right 62.2% of the time. Every model must beat that baseline.

2. Speed and volume separate the classes. High-congestion days show lower average speed and higher traffic volume, with little overlap in the boxplots.
3. Area, day of week, and weather add little. The notebook reads High-congestion rates as roughly uniform across the eight areas, seven days, and weather types, which points to a city-wide pattern. These were judged from charts, not tested.
4. Three columns leak the target. Environmental Impact is an exact linear function of Traffic Volume (r = 1.00). Road Capacity Utilization (r = 0.865) and Travel Time Index (r = 0.752) are measured after congestion occurs. All were removed.
5. Ceiling effects. Road Capacity Utilization and Travel Time Index sit at their maximums (100 and 1.5) in at least half of rows, and Congestion Level reaches 100 in at least a quarter, so the 80% cutoff sits on a heavily capped scale.

3. Problem Statement

Traffic authorities react to gridlock after it forms. This project asks whether historical daily observations of vehicle volume, average speed, incidents, pedestrian and cyclist counts, road, area, weather, roadwork, and calendar position can classify a day at a given road as High Congestion (80% or more) or Moderate/Low, and which of five algorithms does it most reliably.

The EDA shows a 62.2% majority class and strong speed and volume signals, so the study reports each model against the 62.2% always-High baseline instead of accuracy alone. The scope is classification of individual road-day observations, not forecasting of future days, and it does not establish what causes congestion or which intervention would reduce it.

4. Target Audience and Decisions

The intended users are municipal traffic authorities, urban planners, and traffic engineers. The analysis can support these decisions:

- Which conditions (low speed, high volume, incident counts) mark high-risk road-days that deserve earlier intervention such as signal re-timing or advisories.
- Whether a simple model is reliable enough to justify a proactive alerting pilot, or better data is needed first.
- How much weight to give model output: it suits a feasibility discussion, not live operations.

It cannot support decisions that need cause and effect (which fix reduces congestion) or next-day forecasting.

5. Chosen Analytics

The project uses supervised binary classification, with High Congestion (1) as the positive class. Five models are benchmarked on the same stratified 80/20 split and the same preprocessing pipeline, plus an always-High baseline.

Model	Role
Majority baseline (always High)	Floor to beat: 62.2% accuracy

Model	Role
Logistic Regression	Linear, probabilistic baseline
K-Nearest Neighbors	Distance-based, sensitive to scaling
Support Vector Machine (RBF)	Nonlinear margin classifier
Random Forest	Tree ensemble, gives feature importances
Gaussian Naive Bayes	Probabilistic, assumes independent features

Evaluation. Metrics on the 1,788-row test set: accuracy, precision, recall, F1, and ROC-AUC, with confusion matrices and ROC curves. High-congestion recall is the headline metric, since a missed High day is the costly error. Hyperparameters are tuned by cross-validation inside the training set only. The notebook has already imported the metric functions and model classes.

NLP does not apply because there is no text field, and prescriptive analysis would need a measured relationship between actions and congestion, which this data lacks.

6. Scope, Limitations, and Next Steps

Status. Data collection, profiling, EDA, preprocessing, feature selection, the train/test split, and the pipeline are done. Model training, evaluation, the final report, and any dashboard or presentation are still to do.

Phase	Deliverable	Status
1. Data and EDA	Profiling, target, seven EDA charts, correlation analysis	Done
2. Preprocessing	bangalore_traffic_cleaned.csv, 11 features, stratified split, pipeline	Done
3. Modeling	Train the five models and the baseline; tune with cross-validation	Next
4. Evaluation	Metrics table, confusion matrices, ROC curves, feature importances	Next
5. Interpretation	Compare models, recommend one, state caveats	Next
6. Final output	Report, plus dashboard or presentation if required	Next

Limitations to state in the report

- Traffic Volume and Average Speed may be the inputs that produced Congestion Level, so high scores could reflect the label's recipe, not real predictive skill. Test with and without them.
- A random split of consecutive days at the same roads can leak information; a chronological split (train on 2022 to 2023, test on 2024) is a stricter check.
- Weather categories differ between the notebook text (Snow, Storm) and the data (Overcast, Windy); correct the text.
- The 'roughly uniform' and 'noise' claims (59% to 65% by area, |r| below 0.03) are from charts, not tests, and the three noise columns' correlations are not shown in the notebook.
- Findings show association only, and the source's collection method is not documented.